Full production three-component gradient flow: energy-continuation audit

TECT verification-first repository · claim A2-FULL-PRODUCTION-WELLPOSED

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1. Purpose and scope

This v1.2 note closes only `A2-FULL-ENERGY-CONTINUATION-AUDIT`. The scope is the canonical P1 functional on the fixed three-torus, in six real coordinates, with positive regularisers and $\eta_{\text{shell}} = 0$. The v1.1 note already establishes the local $H^2 \to L^2$ Lipschitz property of the full lower-order map. Here the task is the exact energy identity, the Fourier-Galerkin compactness passage, and global continuation.

The historical backend remains a non-variational proxy and is excluded. This note does not settle positive-time smoothing or continuous H^2 dependence. The claim consequently remains T4, not T6.

2. Functional derivative and finite Galerkin identity

Let $H = L^2(\mathbb{T}^3; \mathbb{R}^6)$ and let L be the positive self-adjoint fourth-order operator fixed by P1. Write

$$\mathcal{F}(u) = \frac{1}{2}\langle u, Lu\rangle_H + \Phi(u), \qquad R(u) = Lu + N(u). \tag{2.1}$$

Here the family and lock matrices are included in L, while Φ contains the quartic, sextic, and Class-II terms. The v1.1 real-coordinate calculation gives

$$N : H^2 \longrightarrow H \ \text{ locally Lipschitz}, \qquad D\Phi(u)[h] = \langle N(u), h\rangle_H \quad (u, h \in H^2). \tag{2.2}$$

Let P_n be the real L^2-orthogonal Fourier projection, $V_n = P_n H$, and $u_{0,n} = P_n u_0$. The finite-dimensional ODE is

$$\partial_t u_n = -P_n R(u_n), \qquad u_n(0) = u_{0,n}. \tag{2.3}$$

For $v \in V_n$, $D(\mathcal{F}|_{V_n})(u_n)[v] = \langle P_n R(u_n), v\rangle_H$. Thus (2.3) is the exact real gradient flow of the restricted functional, and the ordinary finite-dimensional chain rule gives

$$\frac{d}{dt}\mathcal{F}(u_n(t)) = \langle R(u_n), -P_n R(u_n)\rangle_H = -\|P_n R(u_n)\|_H^2 = -\|\partial_t u_n\|_H^2. \tag{2.4}$$

The projection in (2.3) is essential: an unprojected residual would not in general remain in V_n.

3. Uniform bounds and compactness

The production coercivity audit gives

$$\mathcal{F}(u) \geq \frac{c_{H^2}}{2}\|u\|_{H^2}^2 + \frac{\gamma}{12}\|u\|_{L^6}^6 - C_Y|\mathbb{T}^3|. \tag{3.1}$$

Here $c_{H^2} > 0$ and $\gamma > 0$. As $P_n u_0 \to u_0$ in H^2 and $\mathcal{F}$ is continuous on H^2, the Galerkin initial energies are uniformly bounded. Equations (2.4)–(3.1) imply, on every finite interval,

$$\sup_n \|u_n\|_{L^\infty(0,T;H^2)} < \infty, \qquad \sup_n \|\partial_t u_n\|_{L^2(0,T;H)} < \infty. \tag{3.2}$$

By (2.2), $N(u_n)$ is bounded in $L^\infty(0,T;H)$. Since P_n commutes with the Fourier multiplier L, (2.3) gives

$$Lu_n = -\partial_t u_n - P_n N(u_n), \qquad \sup_n \|u_n\|_{L^2(0,T;H^4)} < \infty. \tag{3.3}$$

The graph norm of L is equivalent to the H^4 norm by the positive fourth-order coefficient and bounded internal matrices. Apply Aubin–Lions to $H^4 \Subset H^2 \hookrightarrow H$. After a subsequence,

$$u_n \to u \;\text{ strongly in } L^2(0,T;H^2), \qquad u_n \rightharpoonup u \;\text{ in } L^2(0,T;H^4). \tag{3.4}$$

The local Lipschitz property makes $N(u_n) \to N(u)$ strongly in $L^2(0,T;H)$. Passing through (2.3) yields

$$u \in L^2(0,T;H^4) \cap H^1(0,T;H), \qquad \partial_t u = -Lu - N(u). \tag{3.5}$$

The Hilbert interpolation result for (3.5) gives $u \in C([0,T];H^2)$.

4. Infinite-dimensional Class-II chain rule

Passing only a lower-semicontinuous inequality would not suffice. The identity follows from a direct lemma. If $u \in C([0,T];H^2) \cap H^1(0,T;H)$ and Φ, N satisfy (2.2), then

$$\Phi(u(t)) - \Phi(u(s)) = \int_s^t \langle N(u(\tau)), \partial_\tau u(\tau)\rangle_H d\tau. \tag{4.1}$$

Indeed, apply the fundamental theorem of calculus on the H^2 segment from $u(t)$ to $u(t+h)$:

$$\Phi(u(t+h)) - \Phi(u(t)) = \int_0^1 \langle N(u(t) + \theta\delta_h u(t)), \delta_h u(t)\rangle_H d\theta.$$

After integration in time, continuity in H^2, local Lipschitz continuity of N, and $L^2(0,T;H)$ convergence of difference quotients prove (4.1). The v1.1 Class-II expansion supplies (2.2), so this is a true real-gradient chain rule rather than formal complex differentiation.

For u in (3.5), the spectral quadratic-form chain rule gives

$$\frac{d}{dt}\frac{1}{2}\langle u, Lu\rangle_H = \langle Lu, \partial_t u\rangle_H \quad \text{a.e.} \tag{4.2}$$

Combining (4.1), (4.2), and (3.5) proves

$$\boxed{\mathcal{F}(u(t)) + \int_s^t \|\partial_\tau u(\tau)\|_H^2 d\tau = \mathcal{F}(u(s)), \qquad 0 \le s \le t \le T.} \tag{4.3}$$

5. Global continuation

Taking $s = 0$ in (4.3), the coercive inequality (3.1) supplies a uniform H^2 bound on every finite interval from $\mathcal{F}(u_0)$. Hence the local continuation alternative cannot occur in finite time. As the construction works for arbitrary T, the canonical solution extends globally in H^2. Only the hash-pinned canonical functional, zero shell bias, positive regularisers, and the earlier lower-bound/mapping audits are used here.

6. Independent executable audit

The NumPy-only script `codes/foundations/a2_full_production_energy_continuation_audit.py` reconstructs the canonical energy from the manifest without importing the Torch backend. On real orthonormal low-Fourier Galerkin subspaces at grids $4, 6, 8$, and on zero, homogeneous, random, q-shell, and Class-II-active states, it verifies (2.4). The recorded run passes 12/12 assertions with

$$\max \operatorname{relerr}\big(D\mathcal{F}[-\nabla_{V_n}\mathcal{F}], -\|\nabla_{V_n}\mathcal{F}\|_H^2\big) = 7.990691441861486 \times 10^{-13}.$$

The real basis Gram error is $2.220446049250313 \times 10^{-16}$. The zero state correctly gives equality; every nonstationary test state has a positive forward descent margin. These values audit signs, factors, real-pairing normalisation, and the projection convention. Sections 2–5 are the analytic closure basis.

7. Devil's-advocate

- *"$\dot{u}_n = -R(u_n)$ does not remain in V_n."* UPHELD as an invalid Galerkin formulation and fixed: (2.3) uses $-P_nR(u_n)$, giving (2.4).
- *"A limiting inequality is being relabelled an identity."* DISMISSED: (3.5) plus the explicit Class-II chain-rule lemma proves (4.3).
- *"Class-II gradients break compactness."* DISMISSED in the pinned regularised scope: $N : H^2 \to H$ is locally Lipschitz, so (3.4) gives strong $N(u_n)$ convergence while (3.3) supplies the H^4 bound.
- *"The executable silently calls the prior autodiff backend."* DISMISSED: it reconstructs the energy in NumPy and uses the backend hash only as a source-pin check.
- *"This establishes smoothing or T6."* UPHELD as an invalid reading. The separate smoothing/continuous-dependence audit remains open.

8. Result footer

```
Result ID:               A2-FULL-PRODUCTION-WELLPOSED
Precise statement:       The canonical full-production H2 flow has the exact
                         L2 energy identity and coercive global continuation.
Scope:                   Fixed T3, six-real-coordinate C3 field, canonical
                         P1 functional, positive regularisers, eta_shell=0.
Dependencies:            A1-PRODUCTION-FUNCTIONAL-REALISATION; v1.1 mapping audit.
Evidence grade:          ANALYTIC + EXECUTED; T4 proof candidate.
Reproduction command:    python codes/foundations/
                         a2_full_production_energy_continuation_audit.py
Expected output:         12/12 PASS;
                         A2-FULL-ENERGY-CONTINUATION-AUDIT-PASS; exit 0.
Falsification gate:      Failed Galerkin chain rule, H2/H4 compactness passage,
                         Class-II chain rule, or finite H2 blow-up.
Tier before / after:     T4 / T4 (one analytic gate discharged only).
No-overclaim statement:  Not smoothing, continuous-dependence completion,
                         T6, T7, historical proxy, or eta_shell!=0.
Next required action:    Audit continuous H2 dependence and the positive-time
                         fourth-order semigroup bootstrap in A2-FULL-SMOOTHING-AUDIT.
```